Climate change: checked against the laws of physics

You do not have to trust a computer model to know whether climate change is real. You only have to trust one thing: the laws of physics.

The one law nobody can argue with

Energy cannot be created or destroyed. That is not an opinion, a theory, or a scientist's claim. It is the First Law of Thermodynamics. A machine that makes energy from nothing is broken — full stop.

A climate model is a machine for simulating the Earth. If it is working, it must conserve energy. If it leaks heat from nowhere, it is broken, and its predictions should not be trusted.

We checked the models' homework

We took 19 of the world's leading climate models and asked one question: **do they conserve energy?**

12 of the 19 fail. They leak heat — two of them at nearly twice the rate of the energy imbalance they are being used to measure. Six of them also leak mass. Most of the "predictions" the world is shown come from machines that cannot even keep their own books balanced.

The part nobody expected

Now throw away the broken models. Keep only the ones that obey the laws of physics. Ask them how bad it gets.

The answer does not get any less serious.

The models that pass the physics check say:

- **~3 °C** of warming for every doubling of carbon dioxide.
- **4.6 to 5.2 °C** of warming by 2100 if we stay on the current path.
- **Below 2 °C** only if we cut emissions hard.

That is the mainstream scientific answer. And it survived the strictest test we could build — not "do I trust this model," but "does this model obey the laws of physics."

Why this should settle it

You do not have to believe a scientist. You do not have to believe a government. You do not have to believe a model.

You only have to believe that energy cannot be created from nothing.

Accept that one law, and the rest is arithmetic. Most models are broken. The honest ones agree with the science. And the science is now verified — not by authority, but by physics.

What this means

- **"It is just a computer model" is now answered.** Most models fail a basic physics test, and the ones that pass agree with the warning.
- **"The models are too uncertain" is now answered.** Most of the uncertainty comes from the broken models. The honest ones point one way.
- **The fix was never "trust us."** It is "check the physics." And the physics is on the side of the warning.

The fine print (so this is checkable, not just asserted)

The conservation test is the First Law of Thermodynamics applied to each model's pre-industrial control run — a steady state where energy in must equal energy out. The per-model leak is from Irving et al. 2021 (19 CMIP6 models); the projections are the verified model's own SSP runs (Nazarenko et al. 2022). The full technical write-up, with the guard tests, is docs/climate-implications.md. The honest caveat, stated rather than hidden: the strictest gate keeps only one model family (GISS), so "3 °C" is that family's value, not an independent crowd — and it lands on the same answer as the IPCC's own assessment.